



Explainable cross-plant transfer learning for fouling prediction in data-limited pilot-scale membrane bioreactors

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ABSTRACT

Accurate membrane fouling prediction is essential for feedforward control and low-energy operation of membrane bioreactors (MBRs). However, reliable model development for data-limited plants is constrained by sparse measurements of mechanistically informative extracellular polymeric substances (EPS) and soluble microbial products (SMP). Here, we developed an explainable cross-plant transfer learning framework for transmembrane pressure (TMP) prediction in data-limited pilot-scale MBRs. Long short-term memory (LSTM) and extreme gradient boosting (XGBoost) base models were pretrained on three data-richer source MBRs treating domestic wastewater, and fine-tuned with limited data from a domestic target plant. The base models predicted source-domain TMP ($R^2 = 0.87$ and 0.86). SHapley Additive exPlanations (SHAP) identified EPS carbohydrates (EPSc), EPS proteins (EPSp), and SMP proteins (SMPP) as the dominant fouling-related drivers. Direct transfer to the target plant yielded limited accuracy ($R^2 = 0.39$ – 0.41), suggesting that the base models did not fully capture its fouling behavior. In contrast, fine-tuning improved prediction accuracy, and the fine-tuned LSTM (LSTM-FT) reached $R^2 = 0.89$ at a 40% fine-tuning ratio. SHAP and leave-one-feature-out (LOFO) analyses revealed that EPSc remained the critical transferable fouling signal, whereas EPSp became increasingly important after fine-tuning. This shift points to an EPSp-related fouling pathway linked to the plant's Fe-enriched influent. The framework was validated on a pilot-scale MBR treating industrial wastewater, where LSTM-FT achieved $R^2 \approx 0.9$, with SMPP emerging as a critical feature. Collectively, the framework preserves shared EPS-centered fouling information while adaptively recalibrating plant-specific drivers, offering a practical route for mechanism-informed fouling prediction in data-limited MBRs.

1. Introduction

Urbanization and increasing wastewater reuse requirements continue to push wastewater treatment toward intensified processes, which need to deliver stable, high-quality effluent within constrained infrastructure footprints [1]. Membrane bioreactors (MBRs), which integrate biological treatment with membrane separation, have therefore been widely adopted for domestic and industrial applications, due to their superior effluent quality and compact configuration. However, membrane fouling remains the principal obstacle to fully realizing these

advantages [2]. Under constant-flux operation, fouling progressively increases filtration resistance and is reflected by a rise in transmembrane pressure (TMP), which in turn increases aeration and pumping demand, intensifies chemical cleaning requirements, accelerates membrane aging, and raises lifecycle costs [3]. Consequently, MBRs typically consume more energy than conventional activated sludge processes (0.4 – 1.6 vs. 0.3 – 0.8 kWh m⁻³) [4]. Accurate prediction of fouling evolution/TMP is therefore essential for enabling feedforward fouling control and low-energy MBR operation [5,6]. Nevertheless, despite substantial progress in fouling characterization and mitigation,

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proactive fouling management remains difficult in MBRs.

To support proactive management, both mechanistic and data-driven approaches have been explored to predict fouling evolution. Mechanistic and hybrid biological–physical models can enhance interpretability, but they are often challenging to parameterize and recalibrate in real plants [7]. In parallel, machine learning (ML) and deep learning (DL) models have emerged as attractive alternatives, because they can capture nonlinear relationships among operational, water-quality, and biomass-related variables [7–9]. Within this landscape, tree-based ML models (e.g., gradient-boosting families such as Extreme Gradient Boosting (XGBoost)) are attractive for tabular process data, whereas recurrent DL architectures (e.g., long short-term memory (LSTM) networks) are particularly suitable for fouling prediction, because membrane fouling is an intrinsically cumulative and path-dependent process, driven by temporal changes in sludge properties and hydrodynamic conditions [10–12].

Notably, comprehensive and rationally designed inputs for model development are key to achieving accurate AI modeling [13]. However, many existing TMP prediction models, particularly those developed at pilot-scale/full-scale MBRs, are built primarily on routinely monitored “conventional” variables, such as mixed liquor suspended solids (MLSS), mixed liquor volatile suspended solids (MLVSS), hydraulic retention time (HRT), and sludge retention time (SRT). Thus, the current approaches for tracking fouling dynamics in MBRs are often insufficient [14,15]. In contrast, mechanistically informative biochemical descriptors such as extracellular polymeric substances (EPS) and soluble microbial products (SMP) are often missing or severely under-sampled at pilot-scale/full-scale MBRs [15,16]. This is a critical limitation, because extensive evidence shows that EPS and SMP, particularly their proteinaceous and polysaccharide fractions, strongly influence fouling propensity by shaping cake formation, compressibility, and pore-blocking behavior [17], although humic and fulvic acids can also contribute to membrane fouling [18–20]. Collectively, this evidence suggests that explicitly introducing EPS and SMP as model inputs could be both scientifically justified and a promising route to improve TMP prediction and interpretability in MBR plants.

Even when these key biochemical indicators are available, pilot-/full-scale MBR datasets are often still insufficient to support the development of robust site-specific fouling prediction models. In practice, measurements of EPS and SMP fractions are usually sparse and irregular, because they rely on labor-intensive laboratory analyses rather than online sensing. Together with missing values, sensor errors, and outliers in routine operational records and key biochemical datasets, these limitations make it difficult for a single data-limited MBR plant to independently train a high-quality predictive model [21,22]. Most existing ML-based MBR fouling prediction models have also been developed and validated using relatively long or dense plant-specific historical records, leaving their cross-plant generalizability under data-limited conditions insufficiently established [7,8,23]. Nevertheless, different MBR plants often share comparable fouling mechanisms, including cake-layer accumulation, pore blocking, and the contribution of microbial products to membrane fouling. This suggests that transferable models for fouling formation/prediction may exist across MBR plants. From this perspective, transfer learning provides a promising strategy to address both data scarcity and data silos in MBR fouling prediction [24,25]. Specifically, a base model can be pretrained on multi-source data from different MBR plants to learn generalized relationships among operational conditions, EPS/SMP signatures, and TMP evolution. Then it could be fine-tuned using a small amount of data from the target MBR to learn the plant-specific fouling drivers.

However, cross-plant transfer learning for MBR fouling prediction is not a simple model-porting problem. First, source and target MBRs may differ substantially in influent composition, sludge physicochemical properties, EPS/SMP profiles, metal–organic interactions, and operating conditions. Moreover, even after fine-tuning, limited target plant data may still be insufficient for the model to fully capture newly dominant

fouling drivers that were weak or absent in the source plants. Furthermore, the success of transfer learning should not be evaluated by predictive accuracy alone, but also by how and why physically meaningful fouling signals are retained, discarded, or reweighted during transfer. Explainable analyses, such as SHAP and leave-one-feature-out (LOFO) analysis, are needed to determine how key EPS/SMP-related indicators function as transferable fouling drivers or plant-specific recalibrated signals. Prior applications of these methods to membrane-fouling prediction have focused mainly on model-level feature ranking [26]. However, SHAP attributions are inherently model-dependent and lack a ground truth for validation, so high predictive accuracy alone does not ensure that the highlighted features reliably reflect the true fouling behavior [27]. Accordingly, independent physicochemical characterization, including excitation–emission matrix (EEM), liquid chromatography–organic carbon detection (LC-OCD), and foulant analysis, is needed to evaluate the plausibility of these attribution shifts and support their mechanistic interpretation.

To address these gaps, this study proposes an explainable transfer learning strategy for fouling prediction in pilot-scale MBRs under data-limited conditions. Specifically, (1) base models (based on LSTM and XGBoost) were first pretrained on multi-source datasets collected from three pilot-scale MBR plants (source plants) treating domestic wastewater, covering diverse operating conditions, routine biochemical characteristics, and key components in EPS and SMP. We then (2) fine-tuned the base models using the target plant dataset to establish a fouling prediction model for the data-limited target plant (treating domestic wastewater), while quantifying the relationship between fine-tuning ratio and predictive performance and examining how LSTM and XGBoost architectures influenced transfer performance after fine-tuning. Next, (3) the transfer behavior and transferable fouling-related information were examined using explainable analyses, including SHAP and LOFO analyses, together with physicochemical characterization by EEM, LC-OCD, and foulant analysis. These analyses were used to clarify how key fouling-related indicators, particularly EPS and SMP, were retained or reweighted during pretraining and fine-tuning, and provided mechanistic support for these changes. Finally, (4) the transfer learning framework was further examined in a pilot-scale MBR treating industrial wastewater to test whether cross-plant transfer learning remains feasible under a more distinct wastewater background. Collectively, this study aims to provide a practical and explainable route for establishing fouling prediction models in data-limited MBR facilities through cross-plant transfer learning.

2. Materials and methods

2.1. Data collection

A pilot-scale MBR plant treating domestic wastewater in Singapore was selected as the target plant. The system received Fe-enriched influent, with an Fe dosage of approximately 10–30 mg/L. It had a treatment capacity of 24 m³/d and a process configuration comprising pretreatment, a five-pass step-feed bioreactor with anaerobic/oxic zonation, and a membrane tank (Fig. 1(a)). The TMP profile of the target plant during the monitoring period is shown in Fig. 2(a), the membrane module is shown in Fig. 2(b), and the detailed process and operating parameters are summarized in Text S1. The target plant dataset covered the period from September 2021 to June 2022. Owing to the twice-weekly measurement frequency of EPS and SMP, model development was based only on dates with complete matched measurements for all input variables and TMP, yielding 71 valid temporally matched records. This data-limited target plant setting reflected the practical monitoring constraints of pilot-scale MBR operation and provided a realistic scenario for evaluating the proposed cross-plant transfer learning framework.

Three pilot-scale submerged MBR plants treating domestic wastewater in Singapore were selected as source plants, yielding 332 records

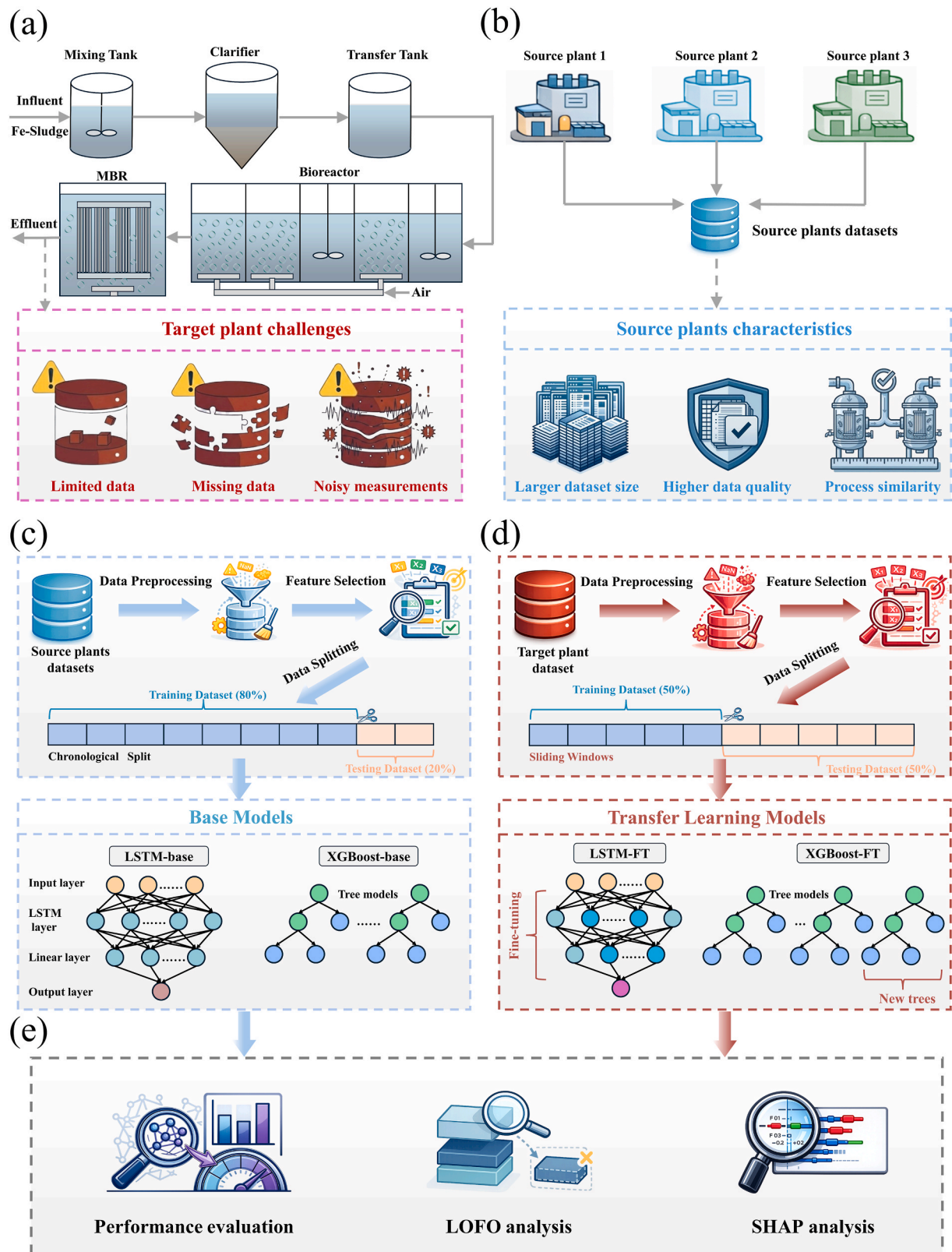


Fig. 1. Framework for transfer learning-based TMP prediction in MBRs: **(a)** treatment process of the target plant, and the associated prediction challenges: limited data, missing data and noisy measurements, **(b)** key characteristics of the three source plants: larger dataset size, higher data quality, and similar process configurations, **(c)** pretraining workflow of the LSTM and XGBoost models using data from the three source plants, **(d)** workflow for fine-tuning the base models using the target plant dataset to obtain the transfer learning models, **(e)** model interpretability and statistical analysis using LOFO and SHAP analyses.

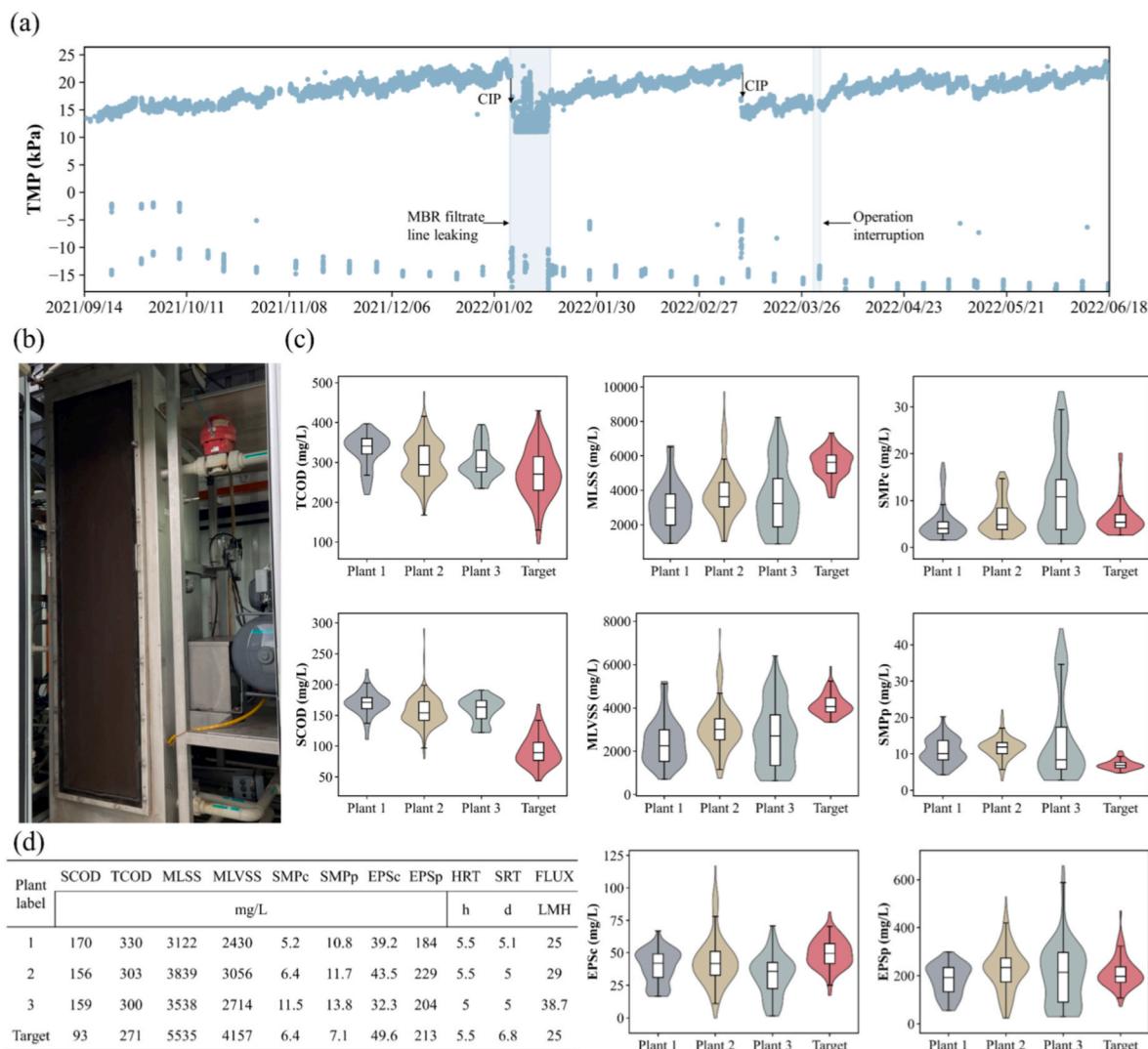


Fig. 2. Operational context of the target MBR plant and source-target dataset characteristics: (a) TMP profile of the target plant during the monitoring period, (b) photograph of the MBR module at the target plant, (c) distributions of the operational and physicochemical variables in the three source plants and the target plant, (d) mean values of these variables for each plant.

in total, including 100 records from Plant 1, 132 records from Plant 2, and 100 records from Plant 3 (Fig. 1(b)). Photographs of the source plants are provided in Fig. S1, and their TMP profiles during the monitoring periods are shown in Fig. S2. These plants had comparable process configurations, consisting of anoxic/oxic biological treatment followed by immersed membrane separation, with conventional sludge recirculation and wasting. Detailed membrane-system characteristics and operating settings are summarized in Table S1. Although the target plant differed in influent composition due to upstream Fe addition, it remained comparable to the source plants in overall treatment configuration, with broadly similar MBR fouling mechanisms, supporting the application of transfer learning between source and target plants.

For each plant, the dataset included 11 variables: HRT, SRT, membrane flux (FLUX), total and soluble chemical oxygen demand (TCOD and SCOD), MLSS, MLVSS, EPSp, EPSc, SMPp, SMPc, and SMPc. TMP was obtained from routine online monitoring and aligned with each sampling date for subsequent analysis.

2.2. Physicochemical analytical methods

HRT, SRT, and FLUX were recorded as routine operational parameters. TCOD and SCOD were determined by the APHA 5220D closed

reflux colorimetric method, and MLSS and MLVSS were measured according to Standard Methods [28]. SMP was defined as the dissolved fraction of the supernatant after 0.45 μ m membrane filtration and EPS was extracted from the residual biomass using the heat method [29]. Proteins and polysaccharides in SMP and EPS, including SMPp, SMPc, EPSp, and EPSc, were quantified using the Modified Lowry Protein Assay and Dubois Method, respectively [30]. Liquid chromatography-organic carbon detection (LC-OCD) was employed to characterize the dissolved organic matter, including biopolymers with molecular weight (MW) greater than 20 kDa and low-molecular-weight (LMW) compounds with MW less than 1000 Da, such as humics, building blocks, LMW acids, and LMW neutrals. The particle size distribution (PSD) of mixed liquor was analyzed using a laser diffraction particle analyzer. Capillary suction time (CST) was measured using a CST apparatus [31]. Total organic carbon (TOC) was measured using a TOC analyzer (TOC-VCSH, Shimadzu, Japan). Excitation-emission matrix (EEM) spectra were measured using a fluorescence spectrophotometer [32]. Total iron was measured using inductively coupled plasma optical emission spectroscopy (ICP-OES).

2.3. Data preprocessing and feature selection

To ensure reliable model development, the raw datasets were first preprocessed by outlier treatment and missing-value imputation (Text S2). To place variables on comparable scales and prevent variables with larger numerical ranges from dominating model training, z-score standardization was applied during preprocessing (Text S3). The scaler was fitted using the training data only, and model outputs were inverse-transformed to the original scale after training.

The final prediction target of this study was next-day TMP, whereas dTMP was used only as an intermediate training target to alleviate non-stationarity and improve output comparability across plants [33,34]. Compared with raw TMP values, dTMP reduced inter-plant differences in TMP level and produced distributions more centered around zero (Fig. S3). Predicted dTMP values were then added to the current-day TMP to reconstruct next-day TMP for model evaluation and presentation.

To assess source–target distribution discrepancy and overlap relevant to transfer learning, joint principal component analysis (PCA) was performed using the input variables together with dTMP. The input and output blocks were standardized separately and assigned equal overall weights (Text S4).

The sample-size-to-feature ratio (SFR) is an important indicator of whether model complexity is appropriate for a given dataset size, and in small-sample modeling an SFR greater than 10 is generally considered desirable [35]. Therefore, Pearson correlation analysis was used as a coarse pre-screening step (Text S5). Because the final prediction target remained next-day TMP, this initial screening was performed against TMP rather than dTMP. It considered both feature–target association and inter-feature collinearity, but served only as a pragmatic dimensionality-control procedure rather than a definitive assessment of nonlinear predictive relevance.

2.4. Construction of base models based on source plants

Given the time-series nature of the data, the source datasets were split within each plant according to chronological order rather than randomly, thereby reducing the risk of information leakage between temporally adjacent samples. Considering the limited sample sizes and short input window, LSTM was selected as a compact recurrent architecture for sequence learning [36]. Transformer- and TCN-based models may offer advantages when larger MBR datasets or longer temporal contexts are available, and systematic benchmarking under such conditions remains a direction for future work [37,38]. Accordingly, two base models were developed: LSTM-base for recurrent sequence learning and XGBoost-base for tree-based learning (Fig. 1(c)). LSTM-base used standardized feature sequences with a sequence length of 3 time steps, whereas XGBoost-base used the same three-step historical information encoded as lagged features. This design ensured that the two base models used the same amount of historical information, allowing a fair comparison of their predictive performance. Additional implementation details are provided in Text S6.

Hyperparameters for both base models were optimized using the training split of the source plant datasets. Because the data were time-ordered, five-fold time-series cross-validation was combined with Optuna-based hyperparameter search to prevent leakage from future samples and improve the robustness of model selection. After optimization, both base models were trained on the same training split, and the trained models were saved to initialize the transfer learning models. The hyperparameter search spaces, final settings, and training details are provided in Table S2–S3 and Text S7.

2.5. Construction of the baseline model for the target plant

Before establishing transfer learning models, a baseline model was developed using only the target plant dataset, and its performance was

compared with that of the transfer learning models. To ensure comparability, the target plant dataset was partitioned in chronological order, with the first 50% used for model training and the final 50% reserved for testing. Under this split, only a limited amount of target plant data was available for model training. With such a small training set, a standalone target-trained LSTM baseline would be more prone to unstable fitting and overfitting, making the resulting comparison less reliable [39]. An independent XGBoost model was therefore adopted as the baseline. It was trained on the training portion of the target plant dataset using the same feature set as the transfer learning models, and the scaler for the target variable was fitted independently using the same training portion. Detailed settings are listed in Table S4.

2.6. Construction of transfer learning models

For transfer learning, the target plant dataset was partitioned chronologically using the same 50:50 split: the first 50% was used as the fine-tuning pool, whereas the final 50% was kept unseen during fine-tuning and reserved as an independent testing set (Fig. 1(d)), thereby reducing the risk of temporal information leakage. The fine-tuning ratio (FT) was defined as the proportion of the total target plant dataset used for fine-tuning, and was examined at 0, 10%, 20%, 30%, 40%, and 50%. The exact numbers of target plant records in the fine-tuning pool, testing set, and each FT setting are summarized in Table S5. The resulting transfer models were denoted as LSTM-FT and XGBoost-FT. At FT = 0, no target plant samples were used for fine-tuning, and the two models therefore corresponded to direct transfer without target plant adaptation.

For each nonzero FT setting, multiple fine-tuning segments of the corresponding length were sampled from the fine-tuning pool to reduce dependence on any single local history (Text S8). Each segment-level experiment was repeated with five random seeds, and performance was reported as the mean $\pm$ standard deviation across all fine-tuning segments and repeated runs. LSTM-FT and XGBoost-FT were fine-tuned using different update strategies, with implementation details and hyperparameter settings provided in Table S6–S7 and Text S9.

All experiments were implemented in Python 3.10.18 using relevant third-party libraries. The deep-learning models were built and trained in PyTorch 2.7.0. The computing environment included an Intel Core i7-14700KF processor (28 logical cores), 32 GB RAM, and an NVIDIA GeForce RTX 5060 Ti GPU.

2.7. Evaluation of model performance

Model performance was evaluated using R^2 , RMSE, and MAE (details in Text S10). Model interpretability was examined using LOFO and SHAP analyses (Fig. 1(e)), as detailed in Text S11 and Text S12, respectively.

2.8. Model adaptability under cross-scenario conditions

Additional validation was conducted using data from a pilot-scale MBR treating mixed petrochemical and pharmaceutical industrial wastewater in Singapore. Detailed process configuration and operating parameters are provided in Table S8. This validation was performed using LSTM-FT under two fine-tuning conditions, FT = 0 and FT = 40%. To maintain consistency with the main analysis, the pretrained model, chronological fine-tuning/testing logic, data preprocessing procedure, and model settings were kept the same. Accordingly, this test was intended to assess whether the established workflow could remain effective under industrial wastewater conditions, without additional model optimization.

3. Results and discussion

3.1. Source–target distribution and PCA analysis

The distributions of the measured variables across the source and target plants are presented in Fig. 2(c), and the corresponding average values are summarized in Fig. 2(d). At the level of core operating variables, the target plant remained within or close to the ranges of the source plants (HRT of 5.5 h versus 5.0–5.5 h, SRT of 6.8 d versus 5.0–7.0 d, and FLUX of 25 LMH versus 25–44 LMH). Across the source plants, the measured variables covered relatively broad ranges of biomass-related and fouling-related properties (e.g., MLSS of 900–9730 mg/L, MLVSS of 639–7671 mg/L, EPSp of 25.0–658.8 mg/L, EPSc of 0.0–117.1 mg/L, SMPc of 0.8–33.3 mg/L, and SMPp of 2.7–44.5 mg/L). By contrast, the target plant showed comparatively narrower distributions for the biomass-related and fouling-related variables (e.g., MLSS of 3580–7340 mg/L, MLVSS of 3340–5920 mg/L,

EPSp of 73.5–469.9 mg/L, EPSc of 17.1–81.3 mg/L, SMPc of 2.6 to 20.1 mg/L, and SMPp of 4.8–10.8 mg/L).

These distributional relationships were further examined by PCA (Fig. 3(a)). The first two principal components explained 74.0% of the total variance, indicating that the major variability in the dataset was captured in a two-dimensional projection. The source plant datasets showed substantial overlap, indicating broadly similar fouling-related variable patterns across the source plants. By contrast, the target plant dataset formed a more compact cluster with partial separation from the source plants, indicating a measurable distribution shift. Importantly, a clear overlap region remained between the target and source datasets. This overlap provided a reasonable basis for developing base models using the source plant datasets and supported the feasibility of subsequent transfer learning.

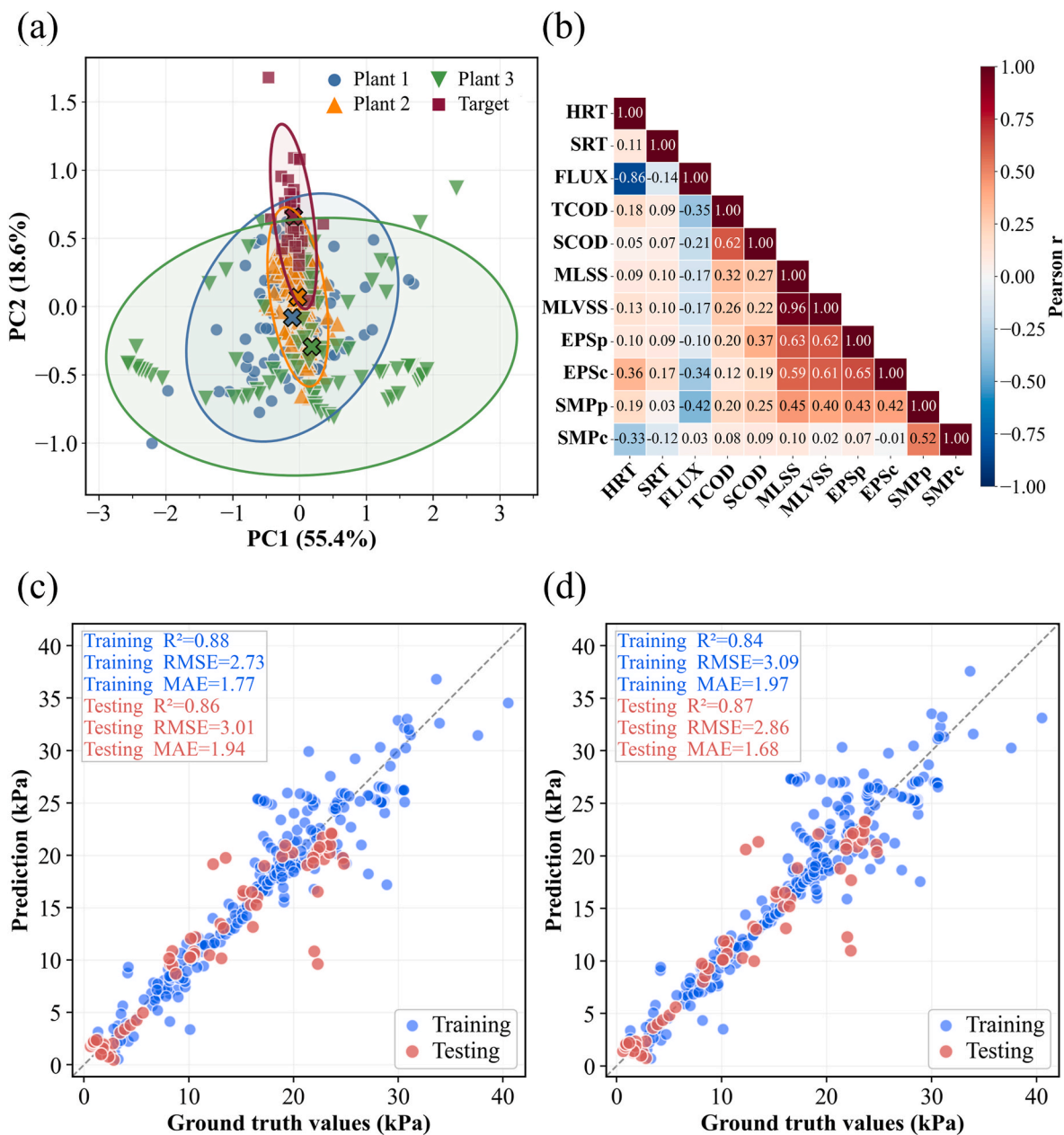


Fig. 3. Source–target data distribution, feature collinearity, and predictive performance of the base models: (a) joint PCA of the source and target plant datasets, (b) feature–feature (X–X) Pearson correlation heatmap, and prediction scatter plots for (c) XGBoost-base and (d) LSTM-base.

3.2. Construction and evaluation of base models

3.2.1. Correlation-based feature selection

Feature–target (X–y) Pearson correlation analysis was further conducted, and the ranking results are shown in Fig. S4. Overall, FLUX, EPSc, HRT, and SMPp showed relatively strong correlations with TMP, with $|r_{x,y}|$ values of 0.471, 0.462, 0.440, and 0.347, respectively. In contrast, SCOD, SMPc, and TCOD showed weak correlations with TMP, with $|r_{x,y}|$ values all below 0.1, and were therefore excluded from the reduced input set used for subsequent modeling. The remaining variables were kept for the subsequent collinearity screening step.

Feature–feature (X–X) Pearson correlation analysis was then performed to further control inter-feature collinearity (Fig. 3(b)). HRT was strongly negatively correlated with FLUX ($r_{x_j,x_k} = -0.86$), likely reflecting their operational coupling: HRT is inversely related to treatment flow rate, whereas FLUX represents the permeate throughput per unit membrane area [8]. MLSS and MLVSS showed a very strong positive correlation ($r_{x_j,x_k} = 0.96$), consistent with MLVSS representing the volatile biomass fraction of MLSS [40]. Within these two collinear feature pairs, FLUX and MLVSS had higher $|r_{x,y}|$ values with TMP and were therefore retained. Accordingly, FLUX, SRT, MLVSS, EPSc, EPSp, and SMPp were retained as a reduced input set for subsequent base model development.

3.2.2. Performance evaluation of base models

The prediction scatter plots for both base models show that most training and testing points were distributed close to the line of equality, indicating good agreement between the predicted and observed TMP values in the source plants. XGBoost-base achieved an R^2 of 0.86, an RMSE of 3.01 kPa, and an MAE of 1.94 kPa on the test set (Fig. 3(c)). LSTM-base showed slightly better performance, with an R^2 of 0.87, an RMSE of 2.86 kPa, and an MAE of 1.68 kPa (Fig. 3(d)). Larger prediction deviations were observed at higher TMP values in both models, suggesting that TMP prediction became more difficult under more severe fouling conditions, possibly due to intensified fouling-layer compression and nonlinear fouling resistance at elevated TMP levels [41]. Overall, both models captured the main TMP variations in the source datasets, with LSTM-base showing a small but consistent advantage over XGBoost-base.

3.2.3. SHAP interpretation of base models and physicochemical characterization

SHAP attribution patterns were further analyzed for both base models. ChatGPT-5.4 was used only as an auxiliary tool to support the summarization of these attribution patterns, while the final interpretation was performed by the authors based on the SHAP plots and corresponding feature-attribution results (Fig. 6(d)). For LSTM-base, the SHAP summary plots showed that EPSc, MLVSS, EPSp, and SMPp made the largest contributions (Fig. 6(a); Fig. S5), whereas FLUX and SRT were comparatively less influential. XGBoost-base showed a much more concentrated attribution pattern, with FLUX dominating the SHAP attribution and contributing more than the other input variables (Fig. S6(a); Fig. S6(b)). These results indicate that the two base models achieved similar predictive performance but relied on different predictive structures during pretraining on the source plants. The broader dependence of LSTM-base on EPSc, MLVSS, EPSp, and SMPp may partly reflect the greater suitability of sequence-learning architectures for variables with stronger temporal continuity and gradual variation [42]. In contrast, the FLUX-dominated attribution pattern in XGBoost-base was consistent with the X–y correlation analysis and may partly reflect the tendency of tree-based models to rely heavily on a dominant predictive signal [43, 44]. These results were consistent with the subsequent transfer learning results, which showed that XGBoost-FT achieved only limited improvement compared with LSTM-FT (discussed in Section 3.3).

Given that LSTM-base achieved strong source-domain predictive

performance and showed much more promise in transfer learning (discussed in Section 3.3), the physicochemical characteristics of biomass in the source plants were further examined to determine whether its learned attribution pattern was consistent with real fouling mechanisms. EPS and SMP were emphasized, because they were identified as important features in the SHAP analysis. Specifically, EPS is more closely associated with sludge flocs and membrane-associated fouling [45], whereas SMP mainly represents soluble organics [46].

Concretely, the three source plants showed a consistent EPS-related signature, including a shared aromatic protein-like peak in the EPS EEM spectra ($E_x/E_m \approx 220\text{--}225/325\text{--}355$ nm; Fig. 4(a)–(c)), and sludge particle-size distributions dominated by particles of 20–40 μm (Fig. 4(d)). These features generally suggested a readily depositable sludge structure that could promote cake-layer formation in MBRs [47]. Moreover, the protein fraction represented by EPSp likely comprises aromatic protein-like components, which could be associated with sludge floc aggregation and the initial adhesion of foulants to the membrane surface [48]. By comparison, EPSc may be more closely related to the gel-forming and resistance-enhancing effects of the polysaccharide matrix, thereby further increasing the hydraulic resistance of the fouling layer [49]. The relatively high attribution of MLVSS (revealed by SHAP) was also physically reasonable, as MLVSS represents the biomass fraction that carries EPS and contributes to sludge deposition on the membrane surface [3].

Additionally, SMP from the three source plants exhibited similar EEM spectral features, with a shared aromatic protein-like fluorescence peak ($E_x/E_m \approx 220\text{--}230/330\text{--}350$ nm; Fig. S7(a)–(c)). These aromatic protein-like SMP components may contribute to fouling development through membrane-surface adsorption or retention within the developing fouling layer, particularly during the early stage of membrane fouling [50,51]. This interpretation was further supported by the LC-OCD results (Fig. 4(e) and (f); Table S9), which showed that SMP in the three source plants was dominated by hydrophilic DOC, with humics and biopolymers as the major fractions. For each LC-OCD fraction, rejection was calculated as the percentage decrease from its concentration in SMP to that in the corresponding effluent, relative to its concentration in SMP. On this basis, biopolymers showed the highest rejection across the three source plants, ranging from 90.0 to 95.3%. These results suggest that SMPp may mainly reflect the aromatic protein-like component within the biopolymer pool that was preferentially retained by the membrane and could therefore participate directly in organic membrane fouling [52].

Overall, the shared physicochemical characteristics of source plants provide clear mechanism-informed support for the EPS- and SMP-related attribution pattern learned by LSTM-base. This indicates that the learned structure was not merely statistical but grounded in real fouling-related properties. In turn, this supports the use of LSTM-base as a reliable pretrained model for subsequent transfer learning.

3.3. Transfer learning performance and interpretation

3.3.1. Effect of fine-tuning ratio on transfer learning performance

The effect of fine-tuning ratio (FT) on transfer learning performance was examined to determine the amount of target-plant data required for effective adaptation. At $FT = 0$, representing direct transfer without target plant adaptation, the pretrained LSTM and XGBoost models showed similarly low R^2 values of 0.41 and 0.39, respectively, indicating limited predictive power under zero-shot transfer. Together with the PCA results showing a measurable source–target distribution shift, this finding suggests that direct transfer alone was insufficient for reliable TMP prediction in the target plant [53].

For LSTM-FT, the largest gain occurred between $FT = 0$ and $FT = 10\%$, with R^2 increasing from 0.41 to 0.81 (Fig. 5(a)). As FT further increased, the performance improved more gradually, reaching R^2 , RMSE, and MAE values of 0.89, 0.50 kPa, and 0.33 kPa, respectively, at $FT = 40\%$ (Fig. 5(b) and (c)). At $FT = 50\%$, the improvement relative to

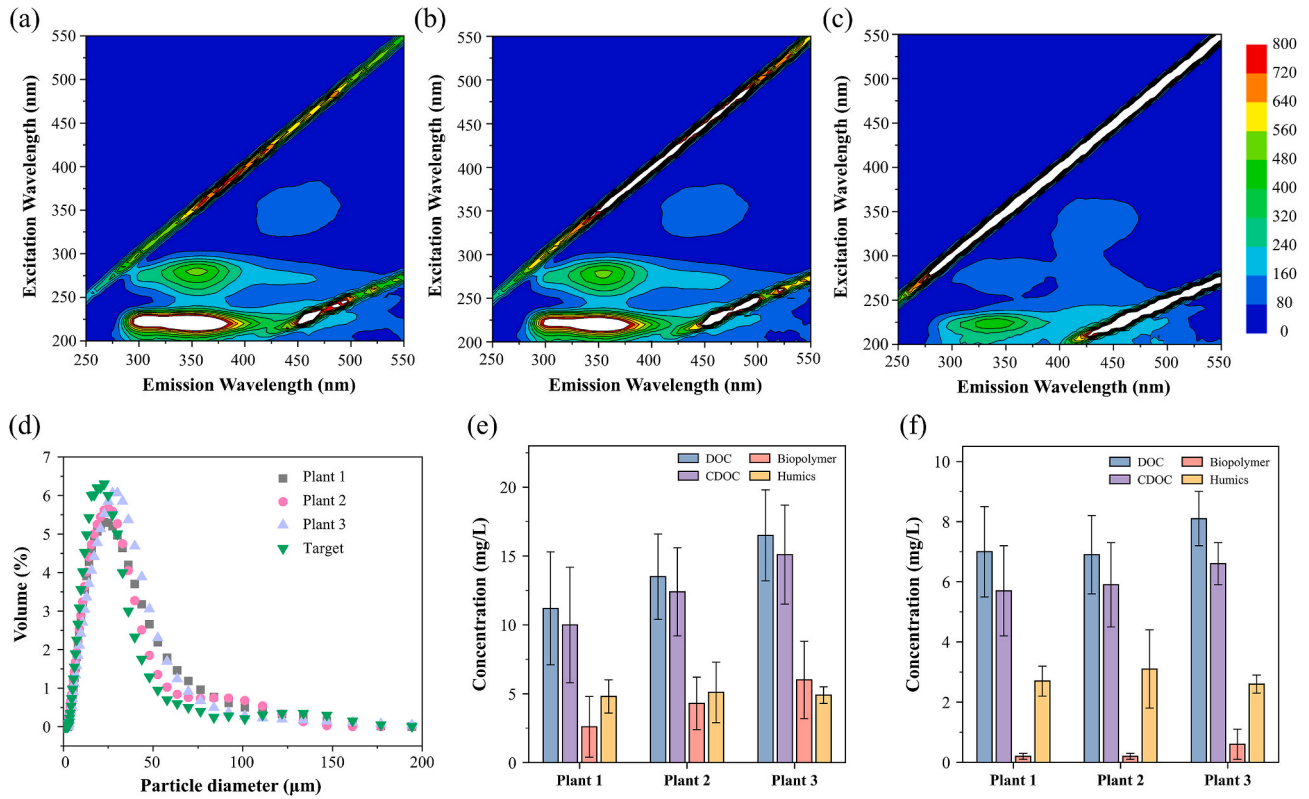


Fig. 4. Physicochemical evidence supporting the EPS- and SMP-related attribution pattern learned by LSTM-base in the source plants: EEM spectra of EPS from (a) Plant 1, (b) Plant 2, and (c) Plant 3; (d) sludge particle-size distributions of Plants 1–3 and the target plant; LC-OCD-derived organic fraction concentrations in (e) SMP and (f) effluent from Plants 1–3.

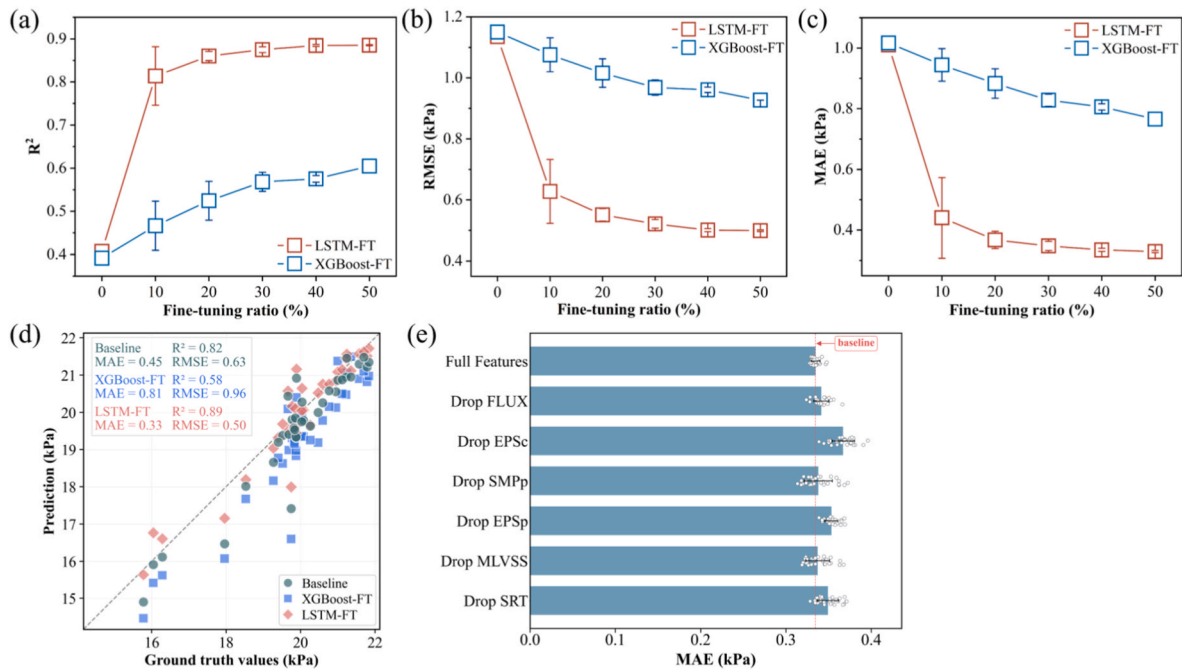


Fig. 5. Effect of fine-tuning ratio on target plant prediction performance and LOFO analysis: (a) R^2 , (b) RMSE, and (c) MAE as functions of fine-tuning ratio, (d) observed versus predicted TMP values for the baseline model, XGBoost-FT, and LSTM-FT on the target plant testing set, with the transfer models evaluated at FT = 40%, and (e) LOFO analysis results for LSTM-FT at FT = 40%. For each nonzero FT setting in (a)–(c), performance is reported as mean $\pm$ standard deviation across all fine-tuning segments and repeated runs, using up to six contiguous segments sampled from the fine-tuning pool and five random seeds for each retained segment.

FT = 40% was only marginal. In comparison, XGBoost-FT also improved with increasing FT, but the gains were smaller and more gradual, with

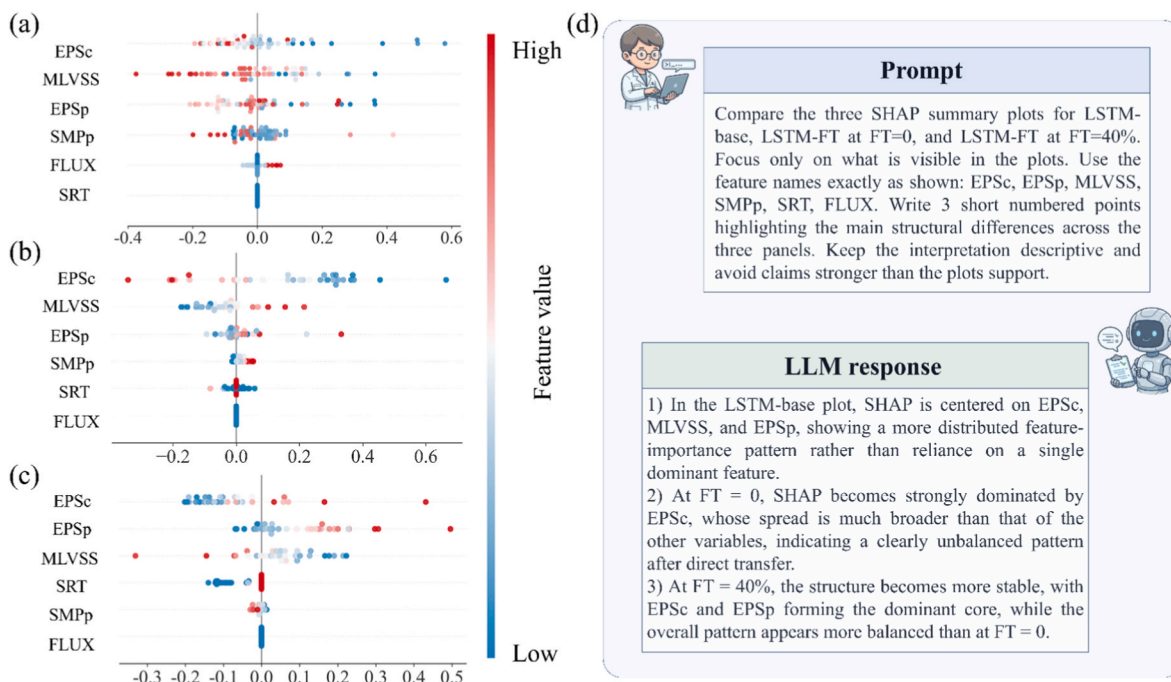


Fig. 6. SHAP attribution patterns and LLM-assisted summary for LSTM-base and LSTM-FT: SHAP summary plots of (a) LSTM-base, (b) LSTM-FT at FT = 0, (c) LSTM-FT at FT = 40%, (d) LLM-assisted summary of the corresponding SHAP observations.

R^2 increasing from 0.39 at FT = 0 to 0.58 at FT = 40% and only slightly further to 0.61 at FT = 50%. Overall, LSTM-FT showed larger performance gains at low FT levels than XGBoost-FT, and increasing FT beyond 40% provided limited additional improvement. Because both transfer models were adapted from models pretrained on the source plant datasets, these performance improvements should be interpreted as target plant adaptation through transfer learning under realistic data-limited conditions, rather than as conventional model development based solely on the 71 target plant records. In addition, for both transfer models the standard deviation across the fine-tuning segments and repeated runs was largest at FT = 10% and decreased at higher FT, becoming small by FT = 40%. This indicates that too few target records reduced the robustness of adaptation, whereas a moderate amount of target data was sufficient for stable performance. Accordingly, FT = 40% was selected as the representative fine-tuning condition for the subsequent analyses.

3.3.2. Performance comparison of baseline model and transfer learning models

To compare predictive performance on the target plant, the baseline model (specifically trained based on target plant datasets) together with LSTM-FT and XGBoost-FT was evaluated (Fig. 5(d)).

The testing samples were mainly distributed within a TMP range of approximately 19–21.5 kPa, with fewer samples at lower TMP values of around 15.5–18 kPa. Within the main TMP range, LSTM-FT showed the tightest clustering around the line of equality, indicating the best agreement between predicted and observed TMP values. The baseline model showed moderate deviations, whereas XGBoost-FT exhibited the largest scatter and the most evident departure from the line of equality. These patterns suggest that transfer learning improved target plant prediction when implemented with LSTM, whereas XGBoost-FT did not provide a performance advantage under the present setting.

The quantitative metrics showed the same trend. Relative to the baseline model, LSTM-FT reduced RMSE from 0.63 to 0.50 kPa and MAE from 0.45 to 0.33 kPa, corresponding to decreases of 20.6% and 26.7%, respectively, while increasing R^2 from 0.82 to 0.89. By contrast, XGBoost-FT increased RMSE to 0.96 kPa and MAE to 0.81 kPa, and

reduced R^2 to 0.58. Taken together, these results indicate that LSTM-FT achieved the best predictive performance under limited target plant data, outperforming both the baseline model and XGBoost-FT. Therefore, the following sections focus on LSTM-FT to further interpret the observed transfer behavior using LOFO and SHAP analyses.

3.3.3. LOFO and SHAP interpretation of feature contributions during transfer

LOFO and SHAP analyses were used to identify the variables that supported transfer performance in LSTM-FT. LOFO analysis was first conducted to evaluate the effect of each input variable on LSTM-FT performance at FT = 40% (Fig. 5(e)). The reference model was the LSTM-FT model trained with all input variables, which achieved an MAE of 0.33 kPa. Each variable was then removed individually, and the resulting MAE was compared with that of the reference model using paired two-sided Wilcoxon signed-rank tests (Table S10). Removing EPSc caused the largest performance loss, increasing MAE to 0.37 kPa ($p \leq 0.001$), indicating that EPSc was the most important variable for maintaining LSTM-FT performance. Removing EPSp or SRT also increased MAE to 0.35 kPa (both $p \leq 0.001$), whereas removing FLUX caused a smaller but still significant increase ($p \leq 0.01$). Overall, the LOFO results confirmed that EPS-related variables, especially EPSc and EPSp, were important for LSTM-FT performance, while SRT and FLUX also showed smaller but statistically significant effects.

SHAP analysis was further used to examine how the feature-attribution pattern evolved with increasing FT. Under zero-shot transfer (FT = 0), EPSc was the top-ranked feature, contributing the largest share of the total attribution, well above MLVSS and EPSp (Fig. 6(b); Fig. S8(a)). This suggests that, under zero-shot transfer, the transferred model retained a strongly EPSc-centered attribution pattern, reflecting limited recalibration before fine-tuning. From FT = 10% to 30%, the attribution pattern became more distributed, with increased attribution to EPSp and more comparable contributions from EPSc, SRT, and MLVSS (Fig. S9–S10). At FT = 40%, EPSc and EPSp were the two top-ranked features and together accounted for more than half of the total attribution (Fig. 6(c); Fig. S8(b)), indicating a more balanced EPS-centered attribution structure after fine-tuning. The feature ranking and

attribution structure at FT = 50% remained largely similar to those at FT = 40% (Fig. S11), consistent with the limited additional performance improvement beyond FT = 40%. Together with the LOFO results, these findings indicate that LSTM-FT performance was mainly supported by EPS-related variables, with EPSc remaining top-ranked and EPSp becoming more prominent after fine-tuning.

3.3.4. Why did transfer learning enable accurate membrane fouling prediction in the target plant?

Because SHAP and LOFO identify model-relevant variables rather than causal mechanisms, the attribution changes were interpreted against independent physicochemical evidence from the target plant. At FT = 0, the attribution pattern remained dominated by EPSc, and the model retained moderate predictive performance ($R^2 = 0.41$), indicating that the EPSc-centered predictive structure learned by the base model was partially applicable to the target plant. The relatively high EPSc concentration in the target plant (49.6 mg/L) further suggests that polysaccharide-related fouling remained important. Elevated polysaccharide content can enhance water retention and reduce sludge dewaterability, thereby promoting the formation of a highly hydrated, compressible, and high-resistance fouling layer. This interpretation was consistent with the relatively high CST (49.45 s) and specific CST (4.36 s/(g/L)) observed in the target plant (Fig. 7(a)). Taken together, these results suggest that EPSc represented a shared membrane-fouling matrix characteristic across plants that remained stably informative during transfer prediction.

Performance improved when FT increased to 40% ($R^2 = 0.89$), and the most notable change in the SHAP results was the rise of EPSp to the second most important feature. This shift suggests that fine-tuning increased model sensitivity to protein-related EPS characteristics in the target plant. The EPS EEM spectrum of the target plant showed a dominant aromatic protein-like peak (Fig. 7(b)), consistent with the protein-related EPS signature observed in the source plants. Of note, it

was hypothesized that the increased importance of EPSp (aromatic proteins) could be due to the Fe-enriched influent in MBRs [54]. To further support this interpretation, two Fe-dosing stages in the target plant were compared (Table S11). Specifically, upstream Fe was dosed into the influent mixing tank before pretreatment at target dosages of 9 and 26 mg/L, allowing Fe accumulation and EPS compositional responses in the MBR system to be examined. As the Fe dosage increased, the sludge-associated Fe content increased from 45 to 101 mg/g-MLVSS, while EPSp increased from 47.96 to 62.30 mg/g-MLVSS. These observations suggest that Fe-enriched conditions were associated with a more protein-enriched EPS composition, which may explain the increased attribution of EPSp after fine-tuning. Higher EPS protein content is generally associated with stronger hydrophobicity and greater floc aggregation, which may favor membrane-surface adhesion and cake-layer densification [46,55]. This interpretation was consistent with the relatively high viscosity and specific viscosity of the target sludge (50.03 mPa s and 9.46 mPa s/(g/L), respectively; Fig. 7(d)). Taken together, the increased SHAP attribution of EPSp suggests that fine-tuning recalibrated the model toward the protein-related EPS characteristics that were more pronounced under the Fe-enriched conditions of the target plant.

The two-step CIP results further supported the association between the Fe-related EPS response and the composition of the retained fouling layer in the target plant. During the citric acid (CA) cleaning stage, total Fe increased sharply from 0.62 to 43.02 mg/L (Fig. 7(e)), indicating the release of substantial Fe-bearing components from the fouling layer. During the subsequent NaClO cleaning stage, TOC increased from 5.19 to 43.22 mg/L (Fig. 7(f)), while total Fe also increased from 1.85 to 11.16 mg/L, suggesting the release of oxidizable organic matter together with Fe-bearing components. Together with the lower SCOD than TCOD in the target plant (93 vs. 271 mg/L), these results suggest that the dominant fouling layer was not formed by dissolved organics alone, but was more closely associated with a sludge-bound organic matrix

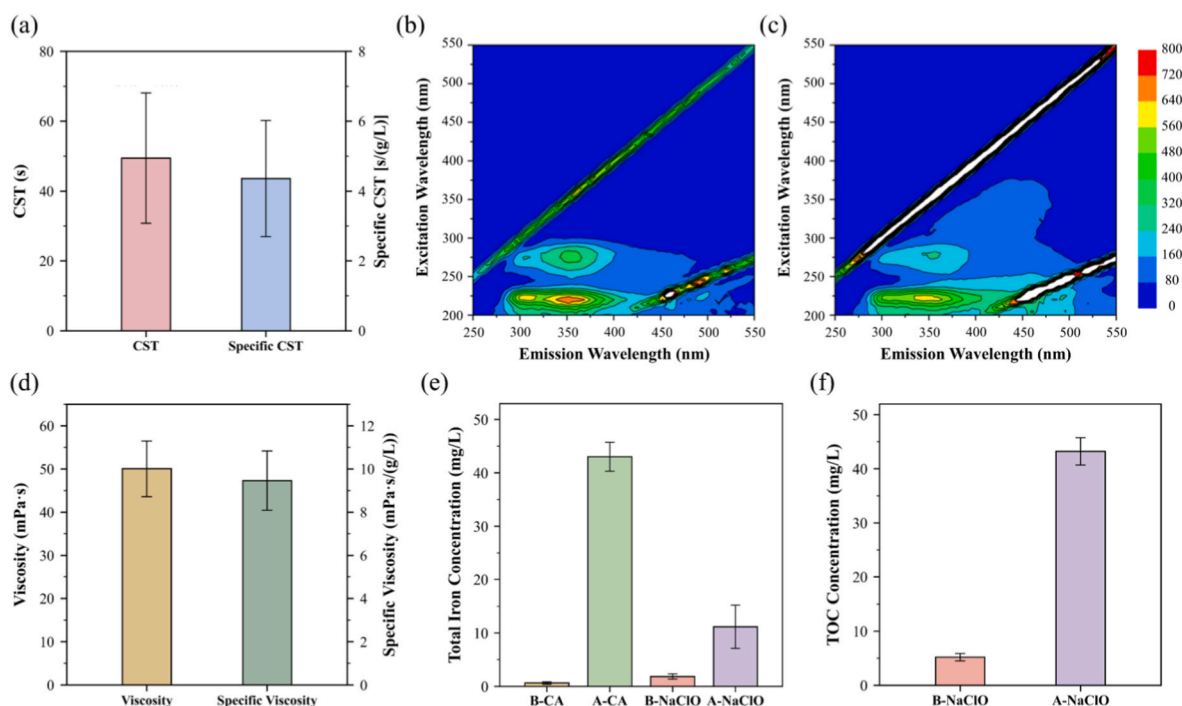


Fig. 7. Physicochemical evidence supporting the EPS-centered attribution pattern learned by LSTM-FT in the target plant: Mixed-liquor properties, including (a) CST and specific CST and (d) viscosity and specific viscosity; EEM spectra of (b) EPS and (c) SMP; (e) total iron concentrations in cleaning solutions collected during the two-step clean-in-place (CIP) procedure, and (f) TOC concentrations in cleaning solutions collected during sodium hypochlorite cleaning. Before CIP, the membrane unit was drained and rinsed with filtrate. The two-step CIP consisted of citric-acid cleaning (15 g/L, pH = 2.5) followed by sodium hypochlorite cleaning (1 g/L, pH = 10), with samples collected immediately before and after each cleaning step. B-CA and A-CA denote samples collected before and after citric-acid cleaning, respectively, whereas B-NaClO and A-NaClO denote samples collected before and after sodium hypochlorite cleaning.

containing Fe-related constituents.

Furthermore, the decline in SMPp attribution after fine-tuning suggests that LSTM-FT placed less emphasis on soluble protein-related information during target plant adaptation. Although the SMP EEM spectrum of the target plant showed a pronounced low-excitation aromatic protein-like feature (Fig. 7(c)), the effluent EEM exhibited broader fluorescence regions ($E_x/E_m \approx 220\text{--}250/300\text{--}460$ nm) that partially overlapped with the low-excitation protein-like region in SMP (Fig. S12). This overlap suggests that part of the low-excitation protein-like dissolved organic signal was not fully retained by the membrane system. Therefore, the reduced attribution of SMPp after fine-tuning does not exclude its contribution to fouling, but suggests that SMPp was less closely associated with the dominant retained foulant layer in the target plant than EPS-related variables.

Taken together, LSTM fine-tuning preserved the transferable EPSc-related fouling information in the target MBR plant, recalibrated the target-specific EPSp-related signal, and placed less emphasis on SMPp, which was less representative of the dominant fouling layer.

3.3.5. Cross-scenario adaptability to an MBR treating industrial wastewater

Because the preceding analysis focused mainly on the domestic MBR scenario, additional validation was conducted using a pilot-scale MBR treating mixed petrochemical and pharmaceutical industrial wastewater. This case provided a preliminary test of the established transfer learning workflow under a more distinct wastewater background. Compared with the domestic target plant, the industrial MBR presented a more challenging transfer scenario, with a more complex influent composition, higher organic loading, and disturbance-driven biomass dynamics. The TMP profile of the industrial MBR showed a clear long-term trend together with multiple abrupt perturbations associated with real operational events (Fig. 8(a)), reflecting complex membrane-fouling behavior under practical engineering conditions.

The industrial MBR differed substantially from the source plants in wastewater and sludge properties, with consistently higher TCOD, SCOD, MLSS, and MLVSS (Fig. 8(b)). Importantly, despite these background differences, EPSc (43.6 ± 20.2 mg/L) remained close to the source plant range ($32.3\text{--}43.5$ mg/L), suggesting that EPSc-related

fouling information was still shared across scenarios. EPSp (256.8 ± 126.7 mg/L) was also of the same order of magnitude as in the source plants ($184\text{--}229$ mg/L), and SMPc (6.9 ± 5.0 mg/L) fell within the source plant range ($5.2\text{--}11.5$ mg/L). These overlaps suggest that part of the fouling-related information learned from the source plants could still be transferred to the industrial MBR, even under a substantially different wastewater background.

Under zero-shot transfer (FT = 0), the base model achieved only moderate performance in the industrial MBR ($R^2 = 0.46$; Fig. 8(c)), suggesting that the fouling-related information learned from the source plants was partially applicable but insufficient to fully capture the TMP dynamics in the industrial MBR. Fine-tuning with industrial MBR data improved prediction performance. This validation used the same chronological fine-tuning/testing logic, pretrained model, preprocessing procedure, and model settings as the main target plant analysis. At FT = 40%, R^2 increased to 0.89, while MAE and RMSE decreased by 68.6% and 54.7%, respectively, relative to FT = 0. This improvement indicates that the established workflow could be adapted to the industrial wastewater background without additional model optimization.

SHAP analysis further showed that EPSc retained the highest attribution after fine-tuning, whereas SMPp increased to the second-highest attribution level (Fig. 8(d)). This pattern differed from that observed in the domestic target MBR plant, where fine-tuning mainly increased the attribution of EPSp. Thus, the two target cases shared stable EPSc-related fouling information, but differed in the feature requiring target-specific recalibration: EPSp in the Fe-enriched domestic MBR and SMPp in the industrial MBR. The increased attribution of SMPp suggests that fine-tuning recalibrated the model toward a more prominent soluble proteinaceous signal under the industrial wastewater background.

The increased attribution of SMPp after fine-tuning was mechanistically plausible in the industrial MBR. SMPp was higher in the industrial MBR (27.6 ± 17.4 mg/L) than in the source plants ($10.8\text{--}13.8$ mg/L), indicating a stronger background of dissolved proteinaceous organics. Previous characterization of this industrial MBR showed pronounced SMP protein peaks during operational disturbances, together with decreases in MLSS and MLVSS, consistent with biomass loss, cell decay, and proteinaceous SMP release into the mixed liquor [56]. Moreover,

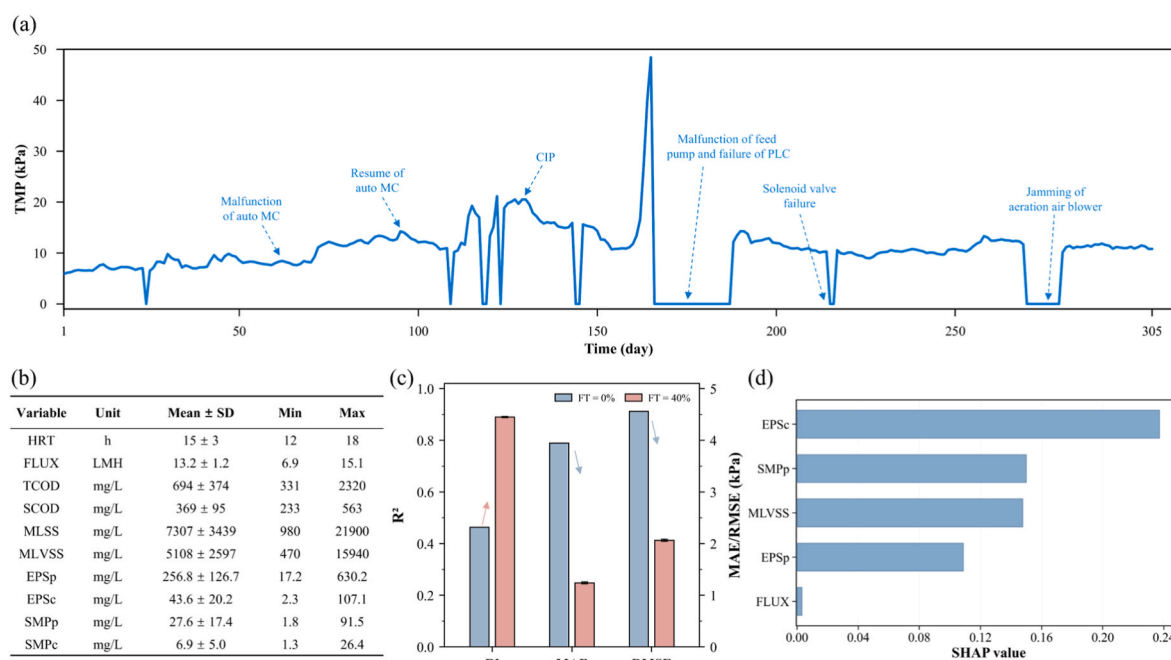


Fig. 8. Additional validation of the proposed transfer learning framework in an industrial wastewater MBR: (a) TMP profile with major operational events, (b) summary statistics of the operational and physicochemical variables in the industrial MBR, (c) the pretrained LSTM model before and after fine-tuning, evaluated at FT = 0 and FT = 40% using R^2 , MAE, and RMSE, and (d) mean absolute SHAP values of LSTM-FT at FT = 40%.

SMP in this MBR was dominated by proteins rather than carbohydrates. These observations suggest that the enhanced SMPp attribution in the transfer model was associated with disturbance-driven soluble protein release. Such proteinaceous SMP could contribute to fouling development, thereby explaining the increased attribution of SMPp after fine-tuning.

Overall, this additional validation suggests that the proposed transfer learning workflow remained applicable to the industrial MBR, not because fouling mechanisms were identical across scenarios, but because the model retained transferable EPSc-related fouling information while fine-tuning recalibrated the attribution structure toward the dominant target-specific signal. In the domestic target MBR plant, this recalibration was reflected by the increased attribution of EPSp under Fe-enriched conditions, whereas in the industrial MBR, it was reflected by the increased attribution of SMPp associated with disturbance-driven soluble protein release. These results further support the ability of the transfer learning framework to preserve transferable fouling-related information while adapting to plant-specific fouling signals across distinct wastewater backgrounds.

4. Implications and outlook

The significance of this study lies not only in improving fouling prediction for a single pilot-scale MBR, but also in providing a basis for more mechanism-informed prediction in data-limited plants. By transferring knowledge learned from data-rich source plants, the LSTM model achieved reliable prediction in a data-limited target plant, suggesting that data-limited plants may not need to build fouling prediction systems entirely from scratch.

More importantly, the present results suggest that indicator selection for fouling prediction should begin with identifying the dominant fouling processes, followed by extraction of representative fouling fingerprints for prediction. Within this framework, indicators such as EPSc and EPSp appear to contain transferable fouling information across MBR plants. However, these variables are expensive, labor-intensive, and difficult to monitor routinely, which limits their direct practical use. A more feasible direction may therefore be to translate key fouling characteristics into monitorable surrogate fingerprints; in this regard, online spectral signals such as UV-vis and fluorescence fingerprints may provide a practical basis for plant-adaptive prediction and proactive fouling control [57].

Nevertheless, several limitations remain and point to directions for future work. The present models were developed from a limited number of pilot-scale MBRs with relatively similar operational backgrounds. At full scale, membrane fouling is further shaped by scale-dependent hydrodynamics and aeration and by greater operational variability than in pilot systems [58], so the pretrained relationships may not directly capture full-scale fouling dynamics. For systems with substantially different influent or operating conditions, the sludge and EPS/SMP composition and the dominant foulants can change, weakening the transferable EPS-centered signal and possibly requiring additional plant-specific indicators for reliable adaptation. The framework should therefore be further validated on larger datasets and extended to full-scale, geographically diverse plants to establish its robustness and generalizability. In addition, the prediction accuracy did not approach unity, indicating that the current indicator set is still incomplete. Besides organic foulants, the target plant also showed evidence of inorganic contributions, particularly iron-associated fouling revealed by the CIP results, which may not yet be fully captured by the present model inputs. Identifying more comprehensive indicators that better capture the combined effects of organic and inorganic foulants may in turn help support earlier fouling warning, better timing of operational adjustment and membrane cleaning, reduced energy and chemical consumption, and improved process stability (Fig. S13).

5. Conclusions

- Reliable base models for MBR fouling prediction were developed by pretraining on the three data-rich source plants. SHAP analysis showed that LSTM-base learned physically meaningful fouling-related patterns mainly through EPSc, EPSp, and SMPp, rather than relying on a single dominant operational signal. This was supported by the shared source plant characteristics, including EPS aromatic protein-like fluorescence, flocculated sludge particle-size distributions, and preferential retention of SMP by the membrane. These results indicate that LSTM-base provided a mechanistically interpretable and reliable pretrained basis for subsequent cross-plant transfer.
- Fine-tuning improved TMP prediction for the target plant treating domestic wastewater, with LSTM-FT achieving $R^2 = 0.89$ at 40% FT. SHAP and LOFO analyses showed that EPSc remained dominant, while EPSp gained importance after fine-tuning. Under the Fe-enriched condition of the target MBR plant, the increased importance of EPSp was associated with a shift toward protein-enriched EPS characteristics. Together, these findings explain why fine-tuning improved prediction accuracy: it retained the transferable EPSc-centered fouling basis while recalibrating the model toward Fe-associated, protein-enriched EPS characteristics that were more closely associated with fouling in the target plant.
- Validation in the industrial MBR further demonstrated that the framework remained effective under a markedly different fouling background. Unlike the target plant treating domestic wastewater, SMPp became the second most important feature in the industrial MBR, reflecting a more prominent proteinaceous SMP signal linked to operational disturbances and biomass decay. The continued dominance of EPSc in the industrial MBR further supported the cross-scenario stability of this transferable fouling basis, while the increased attribution of SMPp showed that fine-tuning could adaptively reweight plant-specific fouling drivers.
- Overall, this study demonstrates that shared fouling-related information can be transferred across MBR plants and selectively recalibrated with limited target-plant data, while independent physicochemical evidence makes this adaptation scientifically interpretable. The framework therefore provides a practical foundation for reliable TMP prediction and plant-adaptive fouling decision support in data-limited MBRs.

CRediT authorship contribution statement

Xiaohang Han: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Liu Yang:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Huan Qin:** Data curation, Investigation. **Shujuan Huang:** Data curation, Investigation. **Han Zhang:** Data curation, Investigation. **Boyan Xu:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing – review & editing. **How Yong Ng:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.memsci.2026.126065>.

Data availability

Data will be made available on request.

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